

$$\min_{u_0 \dots u_{N-1}} \sum_{k=1}^N e_k^\top Q e_k + \sum_{k=0}^{N-1} \tilde{u}_k^\top R \tilde{u}_k + \Delta \tilde{u}_k^\top R_\Delta \Delta \tilde{u}_k$$

$$\text{s. t.} \quad s_{k+1} = \text{PINC}(T, s_k, u_k), \quad u_{\min} \leq u_k \leq u_{\max}$$